Showcase GGPLOT/PLOTLY

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```
library(ggplot2)
mtcars2 <- mtcars
mtcars2$am <- factor(mtcars$am, labels = c('automatic', 'manual'))

ggplot(mtcars2, aes(hp, mpg, color = am)) +
  geom_point() + geom_smooth() +
  theme(legend.position = 'bottom')
```

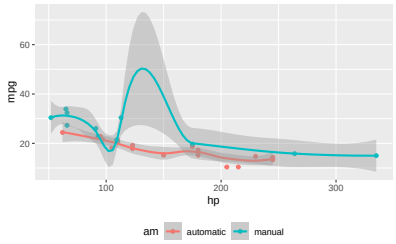


Figure 1: MPG vs horsepower, colored by transmission.

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.1     v readr     2.2.0
v forcats   1.0.1     v stringr   1.6.0
v lubridate 1.9.5     v tibble    3.3.1
v purrr     1.2.2     v tidyr     1.3.2
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-
lib.org/>) to force all conflicts to become errors
```

```
library(sf)
```

```
Linking to GEOS 3.14.1, GDAL 3.12.1, PROJ 9.7.1; sf_use_s2() is TRUE
```

```
library(rnaturalearth)
library(rnaturalearthdata)
```

```
Attaching package: 'rnaturalearthdata'

The following object is masked from 'package:rnaturalearth':

    countries110
```

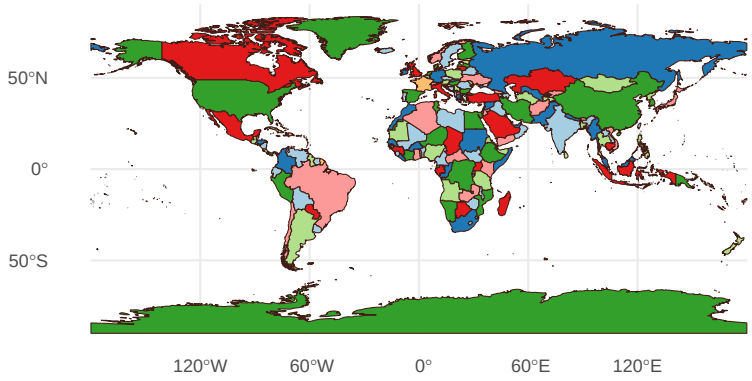
```
library(rnaturalearthhires)

world <- ne_countries(scale = "medium", returnclass = "sf")
class(world)
```

```
[1] "sf"         "data.frame"
```

```
world_without_Australia <- world |>
  filter(iso_a3 != "AUS")

ggplot() +
  geom_sf(data = world_without_Australia,aes(fill = as.factor(mapcolor7)),color = "#401D16", si
  scale_fill_brewer(palette = "Paired") +
  guides(fill = "none") +
  theme_minimal()
```



```
pacific_groups <- ne_download(scale = 50,type = "admin_0_pacific_groupings")
```

```
Reading 'ne_50m_admin_0_pacific_groupings.zip' from naturalearth...
```

```
ggplot() +
  geom_sf(data = pacific_groups)+
  geom_sf(data = world) +
  geom_sf_text(data = world, aes(label = admin))+
  coord_sf(crs = "+proj=ortho +lat_0=0 +lon_0=-160 ") +
  coord_sf(xlim = c(145, 175), ylim = c(-20, 5), expand = FALSE) +
  ggtitle("Solomon Islands", subtitle = "Showing Pacific Island Groupings")+
  theme_void()
```

```
Coordinate system already present.
i Adding new coordinate system, which will replace the existing one.
```

```
Warning in st_point_on_surface.sfc(sf::st_zm(x)): st_point_on_surface may not
give correct results for longitude/latitude data
```

